



CPG to Actionable Care Plans

Bringing actionable intelligence to
healthcare on Red Hat AI

Sam Schiffman



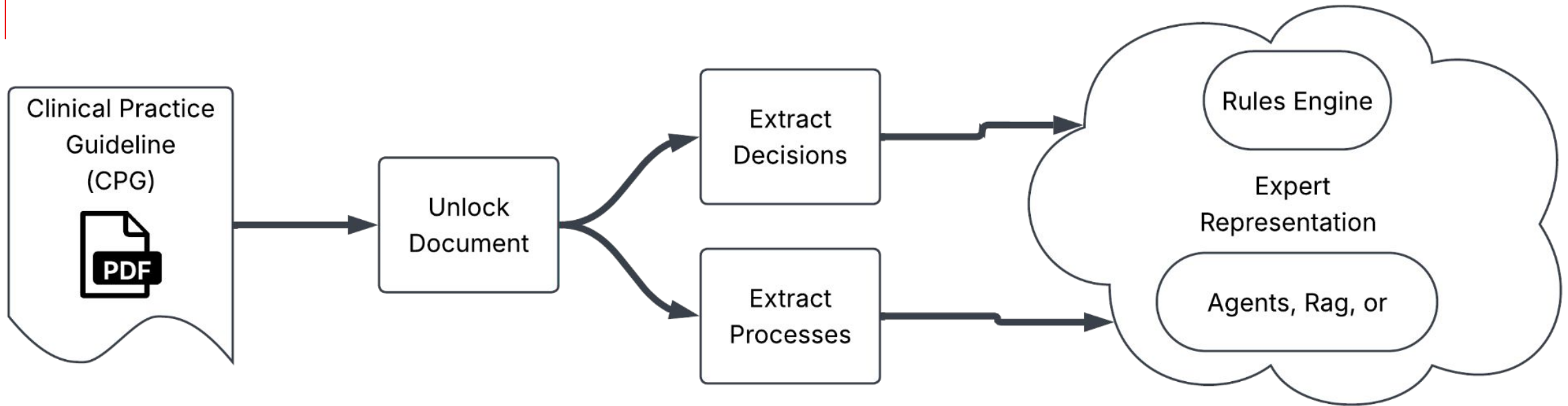
Imagine a patient comes into the Emergency Room. The doctor only has minutes to put together their history and diagnose the current problem. Then they have to build an effective plan for the patient's care.

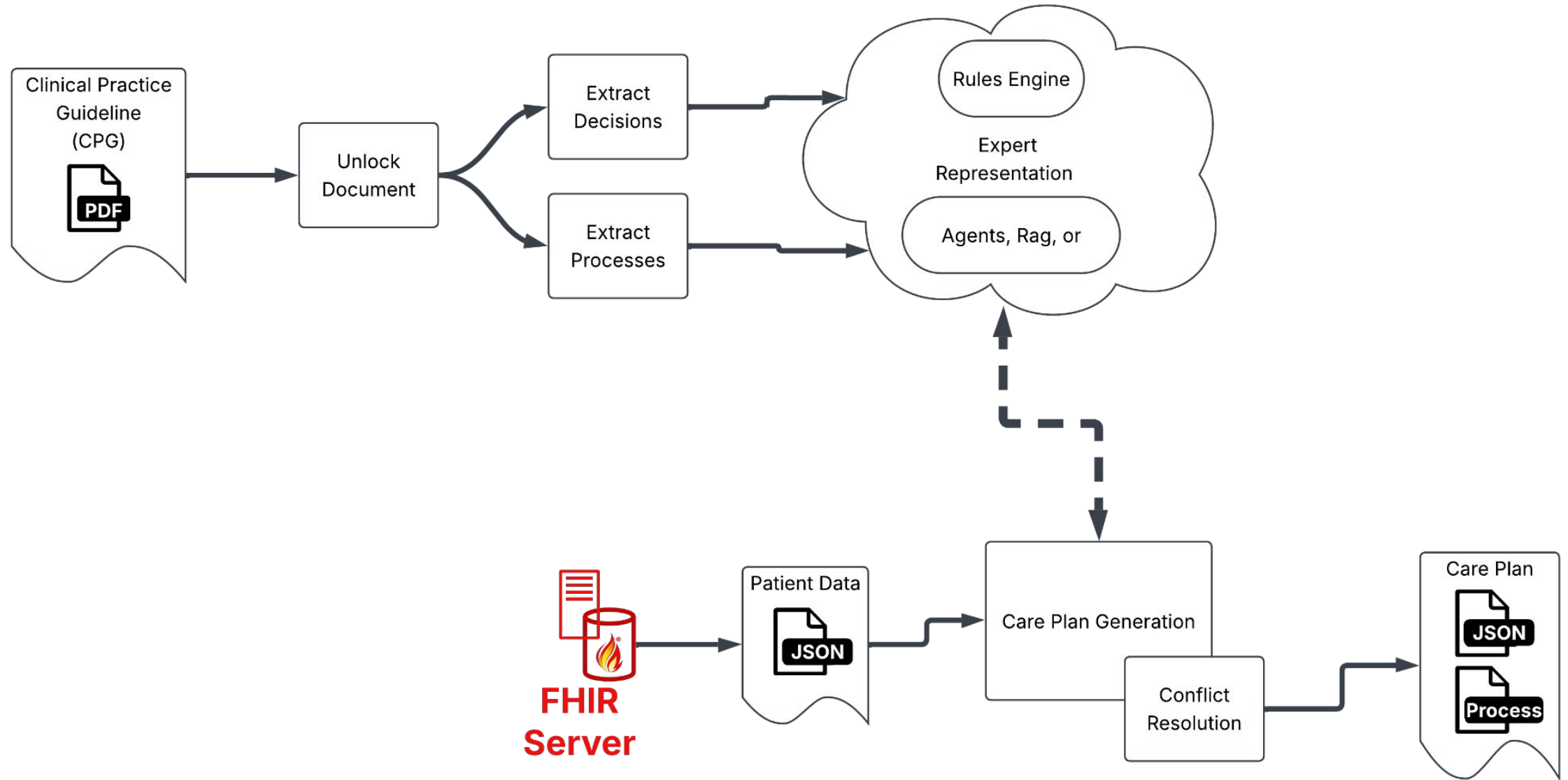
What if we could bring teams of experts right to the doctor's side? Suggest to a doctor what the care plan might be, based on the highest standard of care?

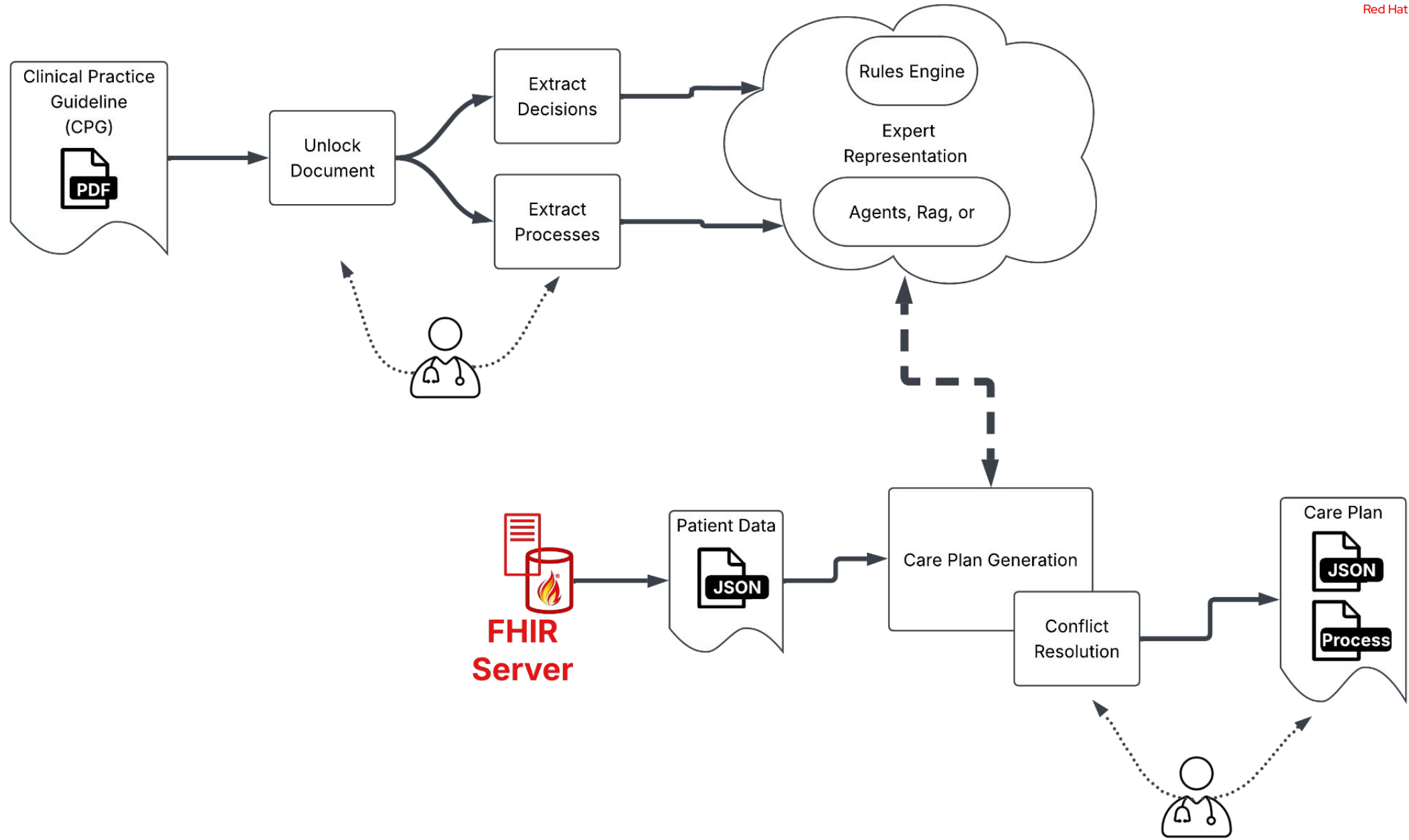
But beyond that, what if the suggestions include automations that could handle appropriate portions of the care?

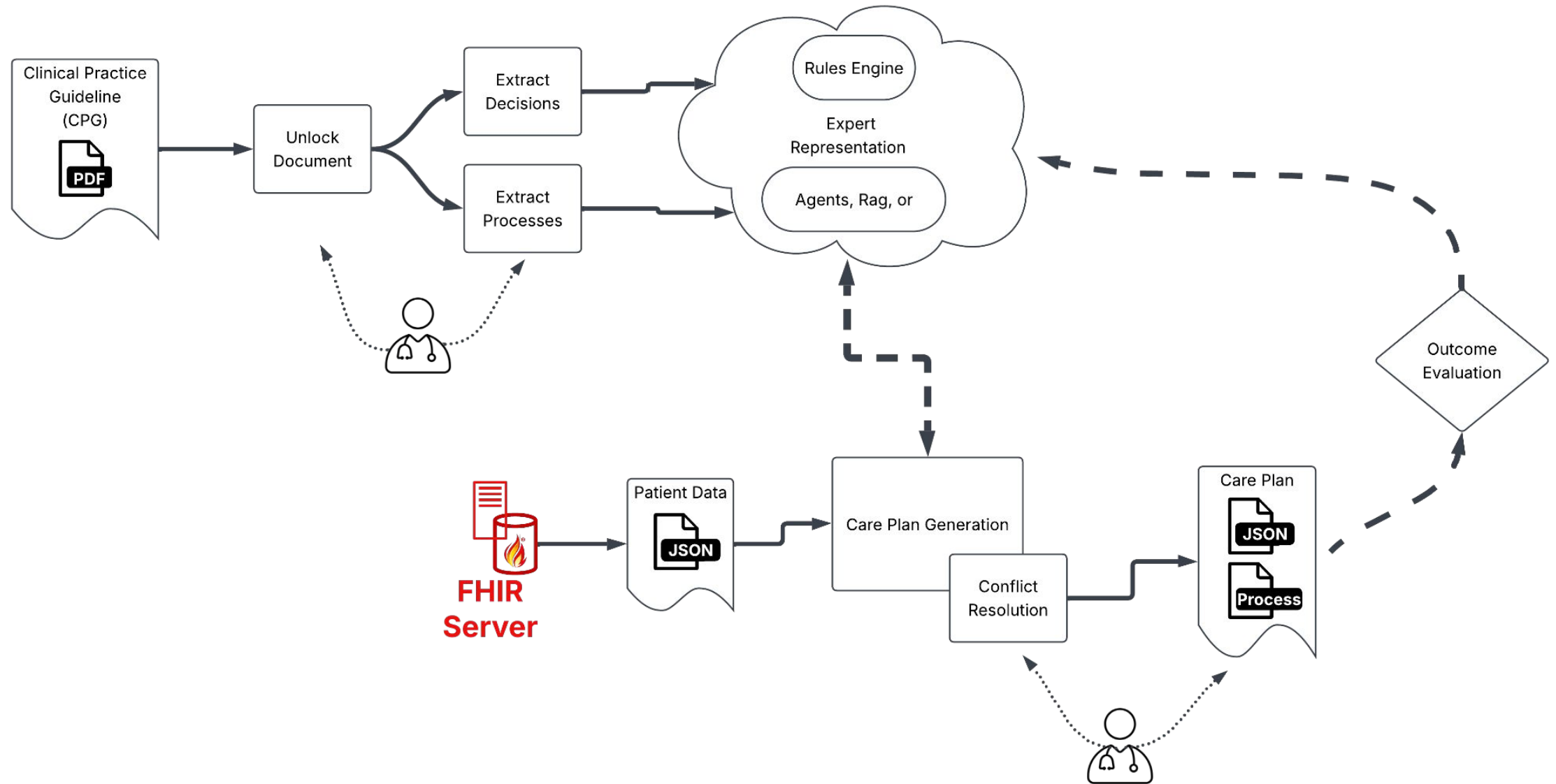
This would be an Actionable Care Plan the doctor could modify, approve, and put into motion right away.

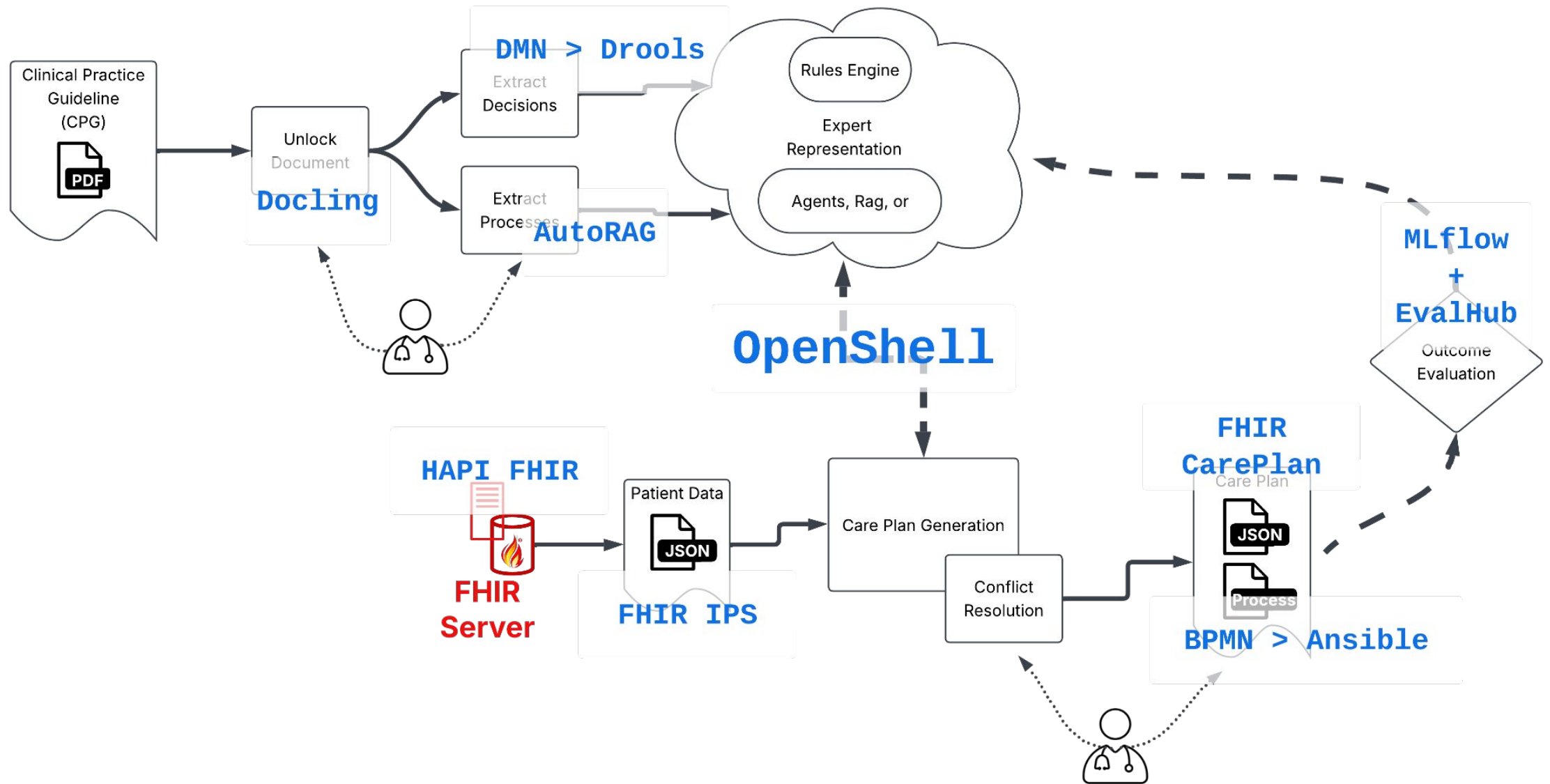




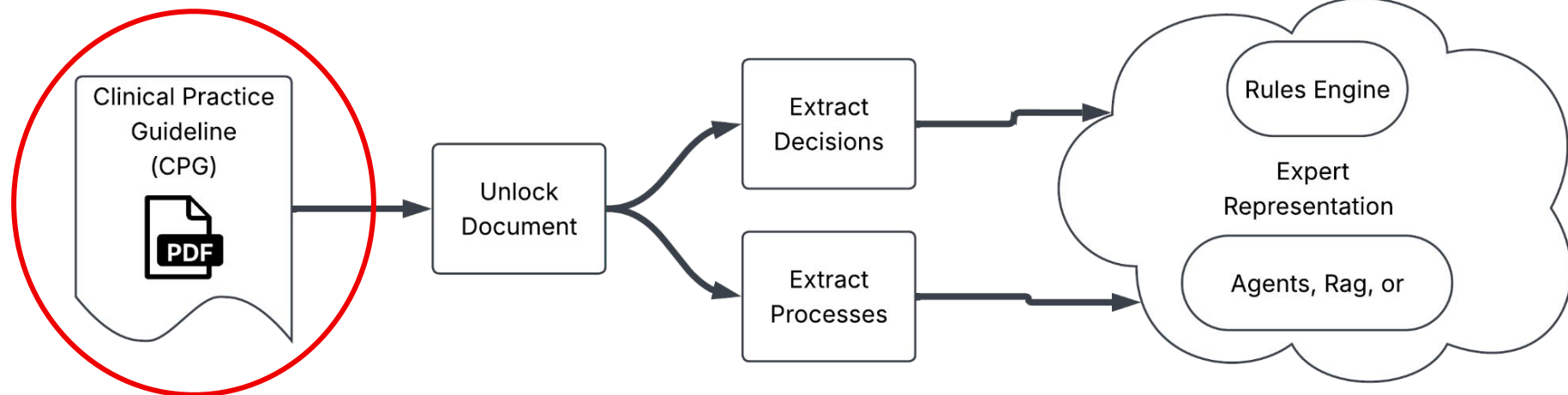






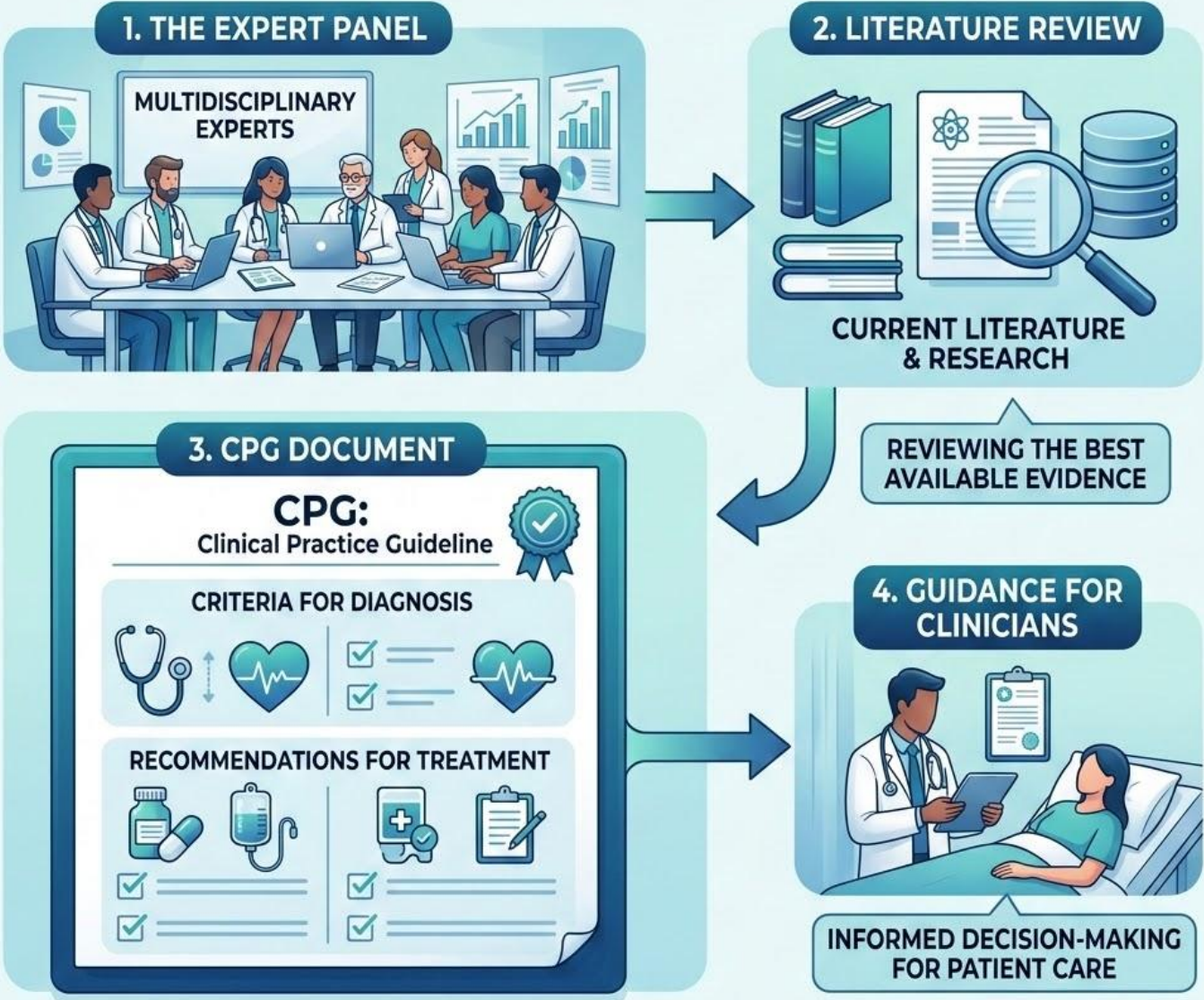


What is a Clinical Practice Guideline (CPG)



WHAT IS A CLINICAL PRACTICE GUIDELINE (CPG)?

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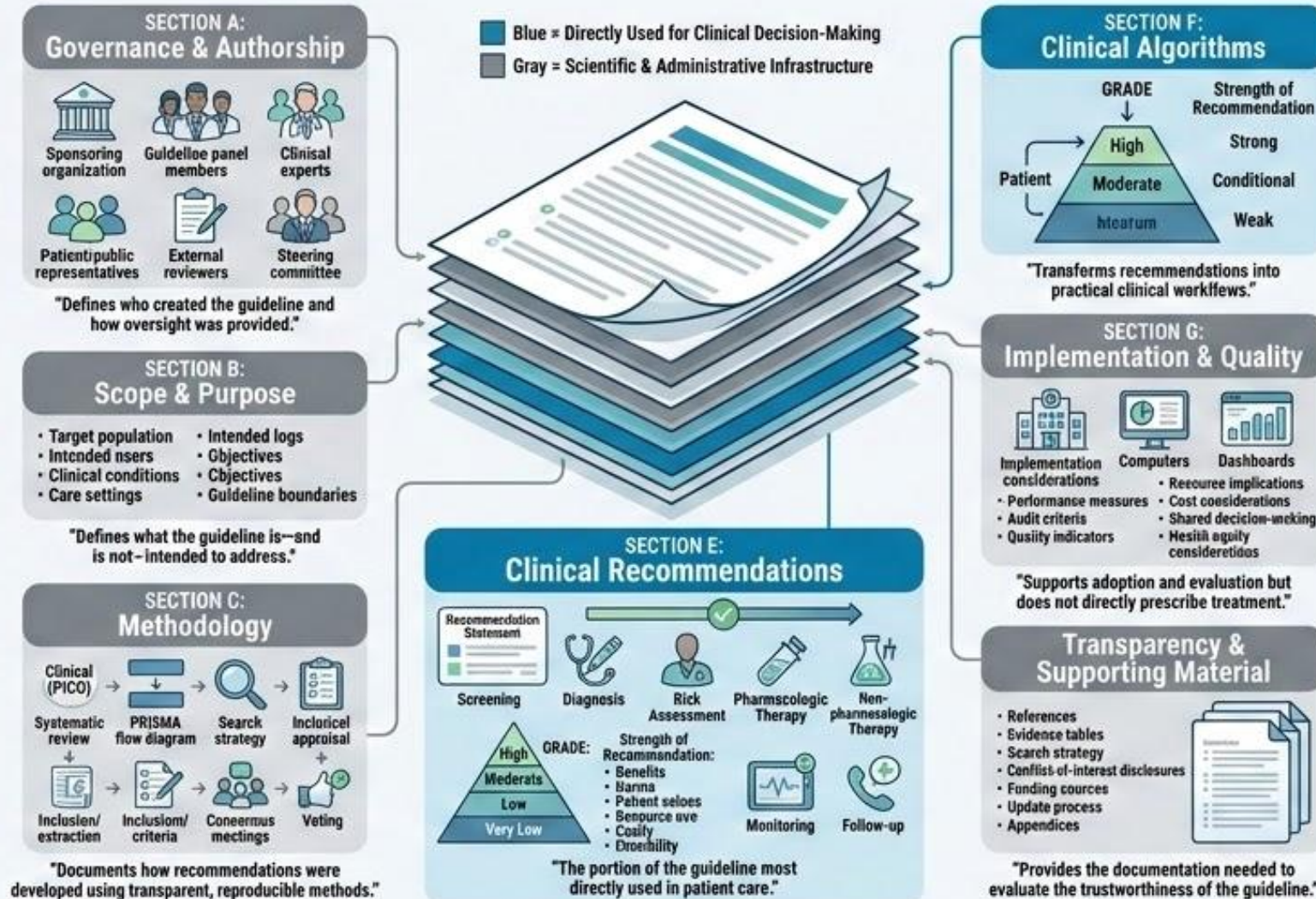
A CLINICAL PRACTICE GUIDELINE (CPG) CAPTURES THE CRITERIA FOR DIAGNOSIS AND RECOMMENDATIONS FOR TREATMENT FOR A MEDICAL CONDITION



Inside a Clinical Practice Guideline (CPG)

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A Clinical Practice Guideline is more than a list of treatment recommendations. It combines **evidence synthesis**, **expert governance**, **transparent methodology**, and **clinical guidance** into a single structured document.



Freely Available Clinical Practice Guidelines (CPGs)

Access authoritative, open-access medical recommendations and structured logic engines from global health systems.

US Dept. of Veterans Affairs

The VA/DoD publishes evidence-based Clinical Practice Guidelines to reduce variation in military and veteran health outcomes.

- Chronic Disease Management
- Mental Health & PTSD Care
- Rehabilitation & Pain Therapy

healthquality.va.gov ↗

World Health Organization

WHO guidelines provide global recommendations on public health interventions, disease prevention, and health system standards.

- Global Health Standards
- Infectious Disease Protocols
- Maternal & Child Health

who.int/publications/who-guidelines ↗

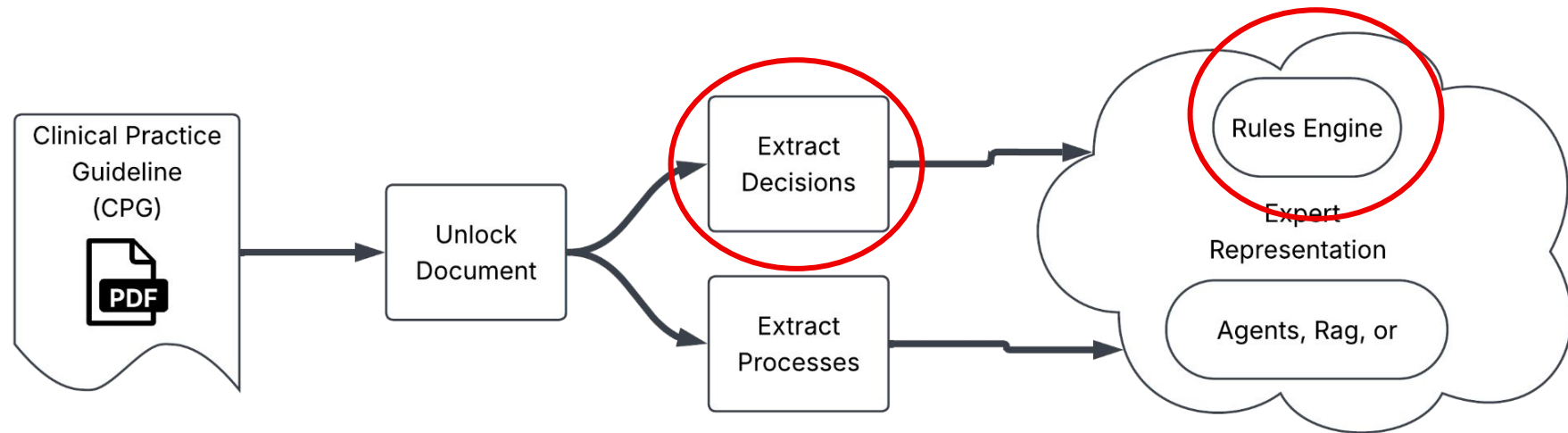
American College of Physicians

The American College of Physicians is the largest medical specialty organization in the United States with members in more than 172 countries worldwide. Our 163,000 members include internal medicine physicians, related subspecialists, and medical students.

acponline.org/clinical-information/clinical-guidelines-recommendations ↗



Extracting Decisions



What is Decision Model and Notation (DMN)?

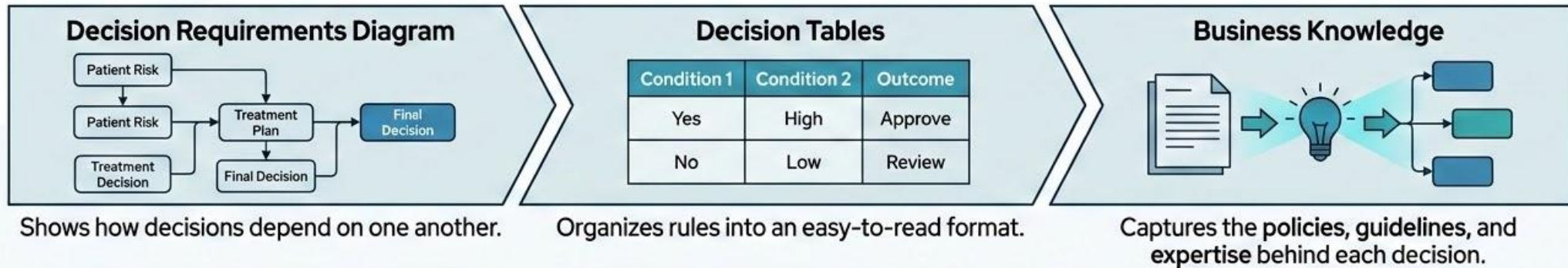
A visual language for describing decisions and the rules behind them.

Why DMN?



DMN transforms complex written rules into clear visual models that are easier to review and understand.

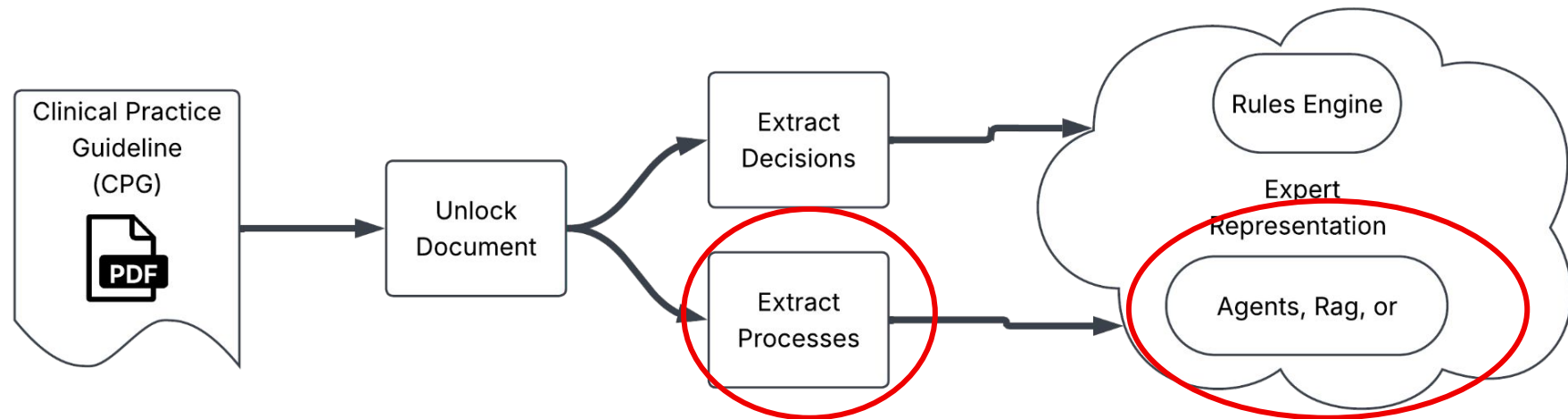
The Building Blocks



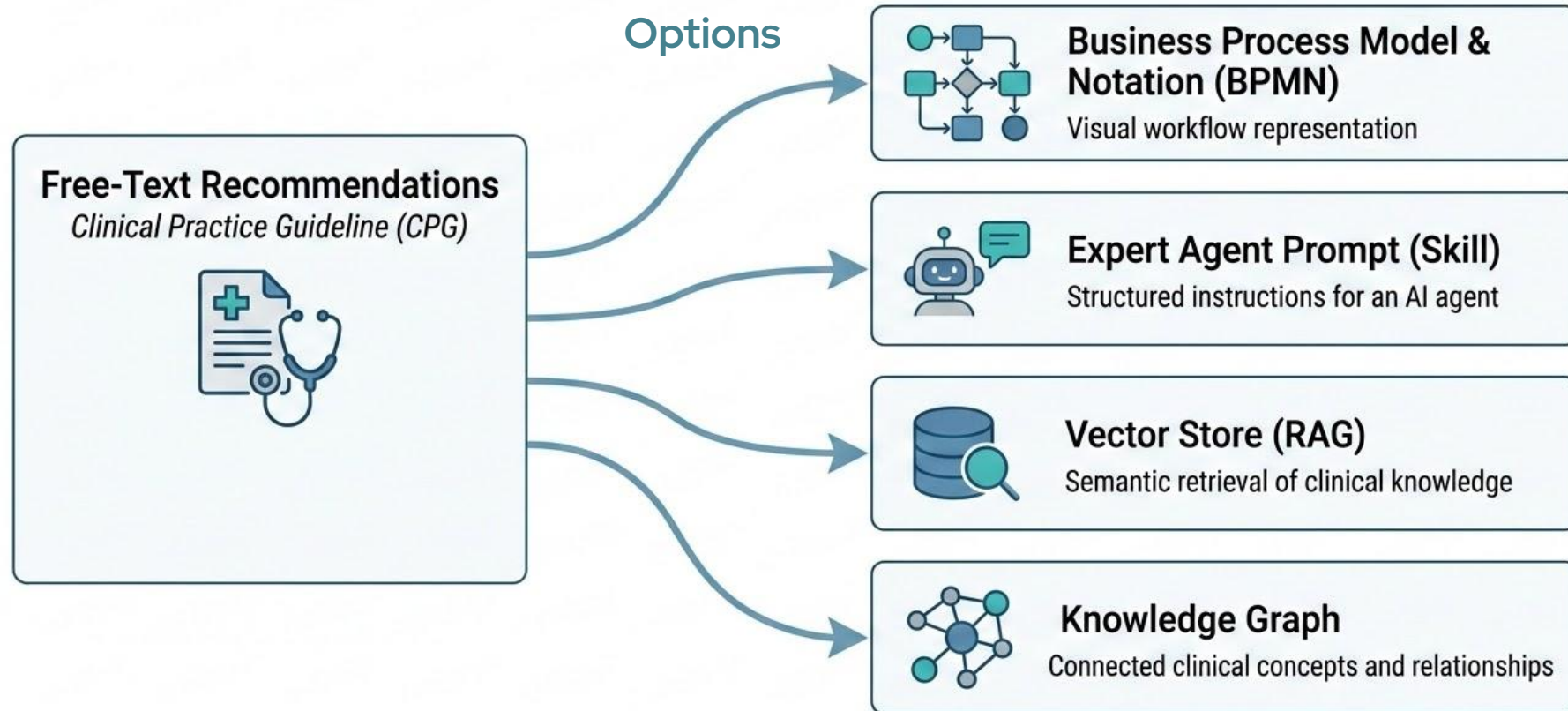
DMN provides a common visual language that helps subject matter experts, analysts, and developers review decision logic together—making complex decisions easier to understand, validate, and maintain.



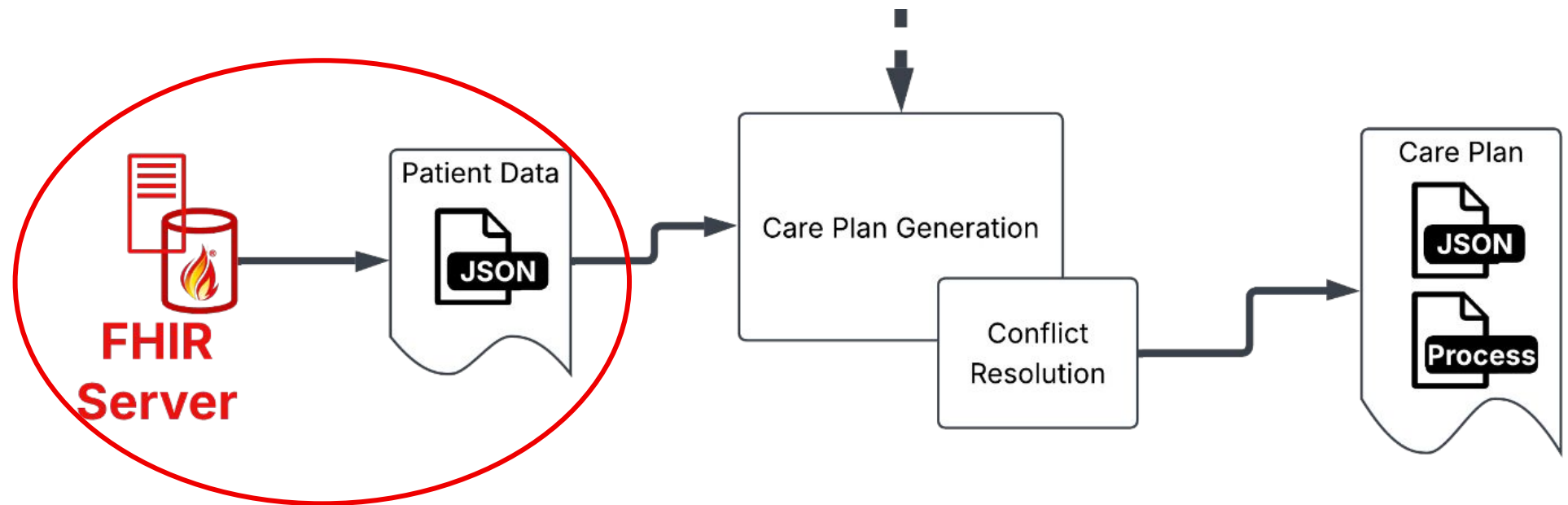
Extracting Process



Transforming Clinical Practice Guideline (CPG) Free-Text Recommendations

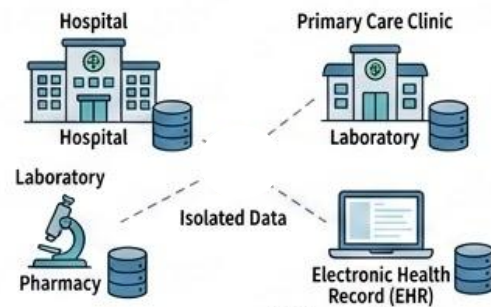


Patient Data



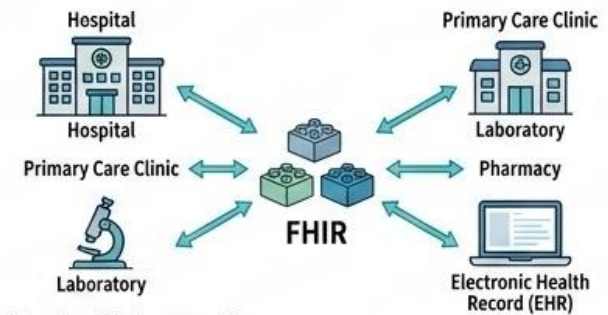
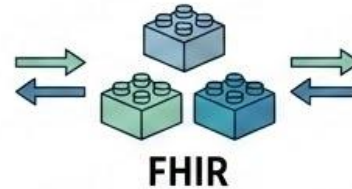
Fast Healthcare Interoperability Resources (FHIR)

A modern standard for sharing healthcare information across systems.

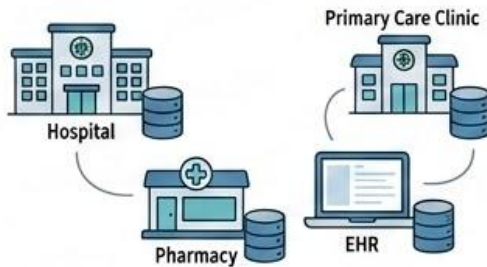


FHIR provides a common standard that allows healthcare systems to exchange information in a consistent and interoperable way.

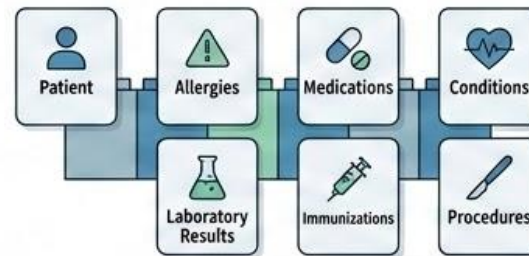
1 Why FHIR?



2 FHIR Organizes Healthcare Information into Resources

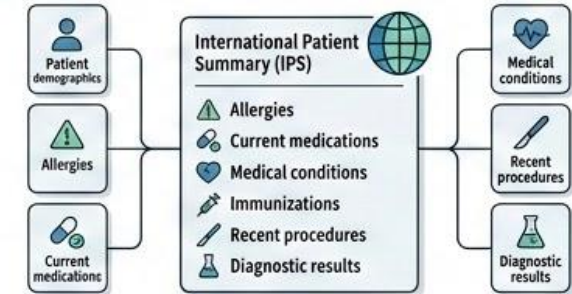


FHIR provides a common standard that allows healthcare systems to exchange information in a consistent and interoperable way.



FHIR organizes healthcare information into reusable "resources" that can be combined to describe a patient's health record.

3 International Patient Summary (IPS)



The International Patient Summary (IPS) is a standardized collection of essential health information designed to support safe patient care across organizations and countries.

Healthcare Systems

FHIR Resources

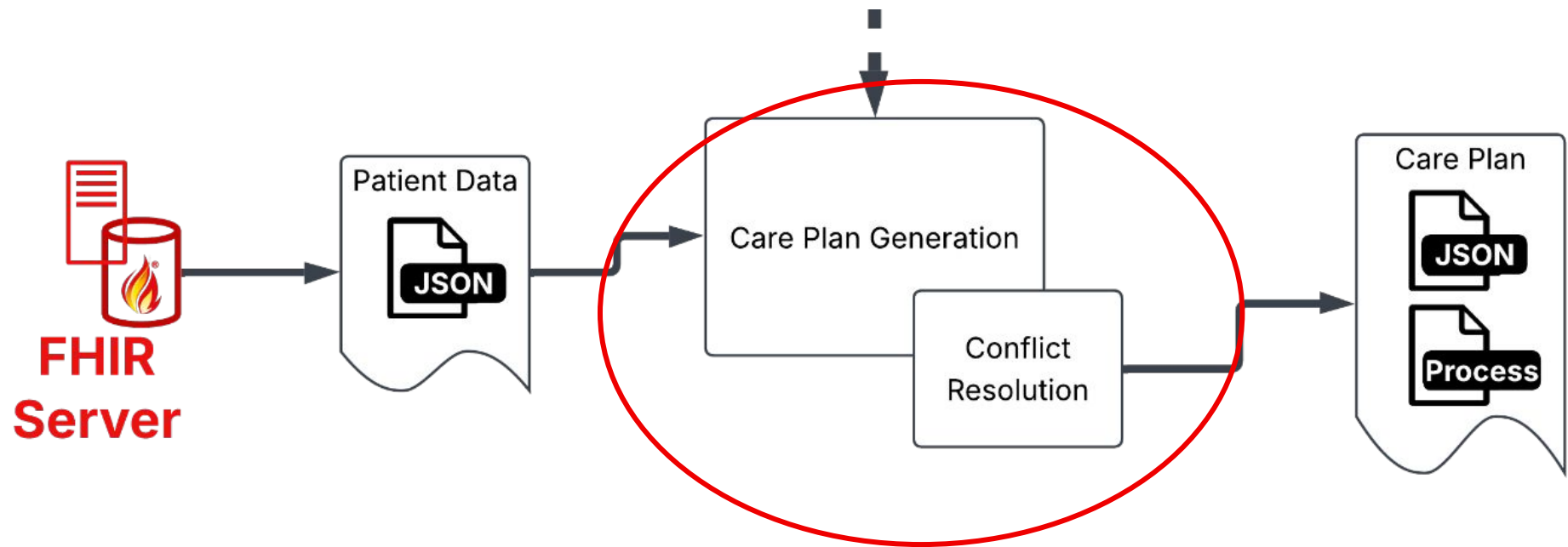
International Patient Summary (IPS)

Shared Clinical Understanding

FHIR provides the common language for healthcare data, while the International Patient Summary (IPS) defines a standardized clinical summary that can be shared wherever a patient receives care.

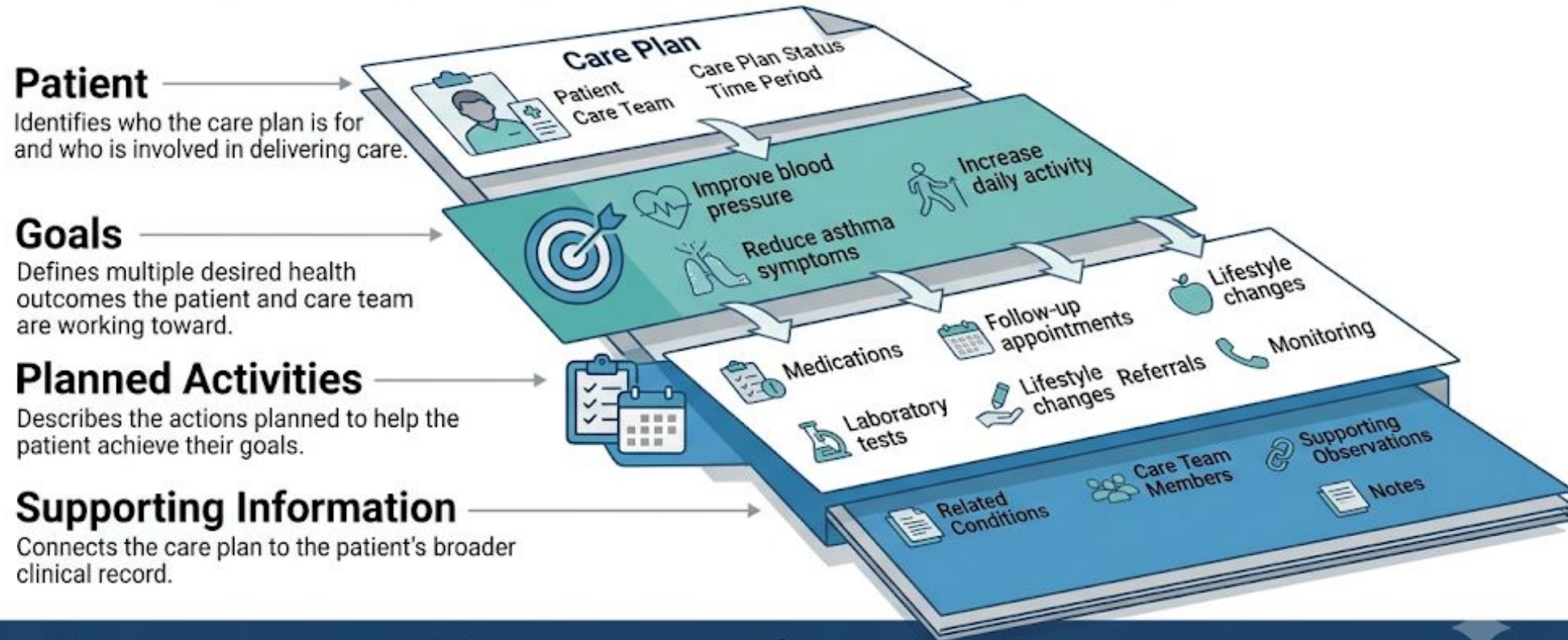


Writing Care Plans



Inside a FHIR Care Plan

A structured plan that organizes patient-centered care around goals and planned activities.



Patient



Goals



Activities

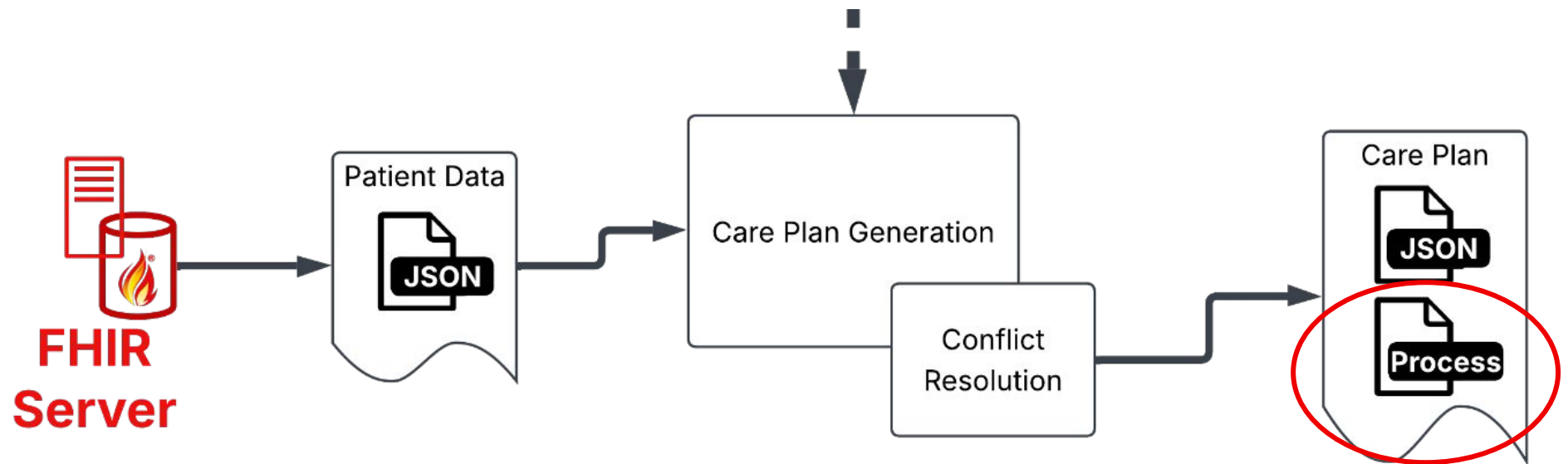


Better Health Outcomes

A FHIR CarePlan provides a standardized, shareable way to coordinate care by linking a patient's goals with the activities needed to achieve them.



Process Automation



Business Process Model and Notation (BPMN)

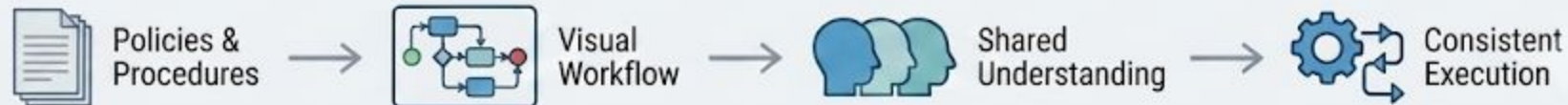
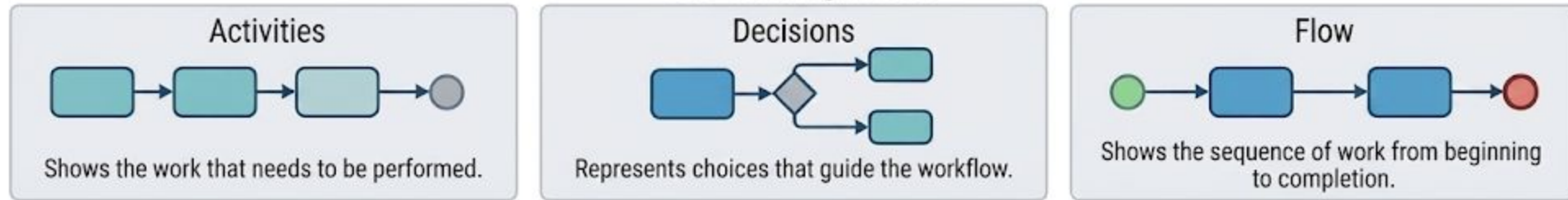
A visual language for describing workflows and business processes.

Why BPMN?



BPMN transforms complex written procedures into clear visual workflows that are easier to understand, review, and improve.

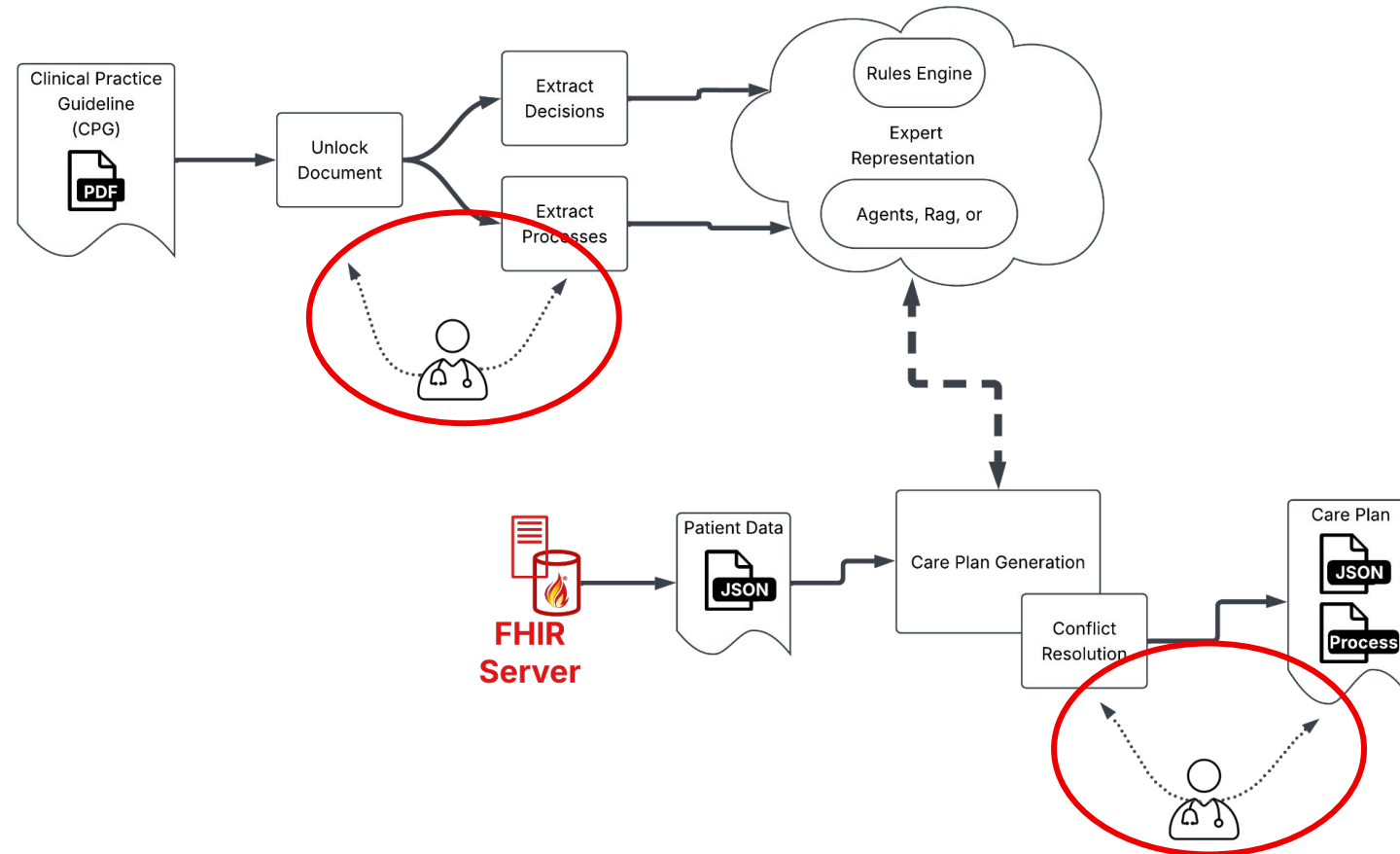
The Building Blocks



BPMN provides a common visual language that helps subject matter experts, analysts, and developers review workflows together—making complex processes easier to understand, improve, and automate.



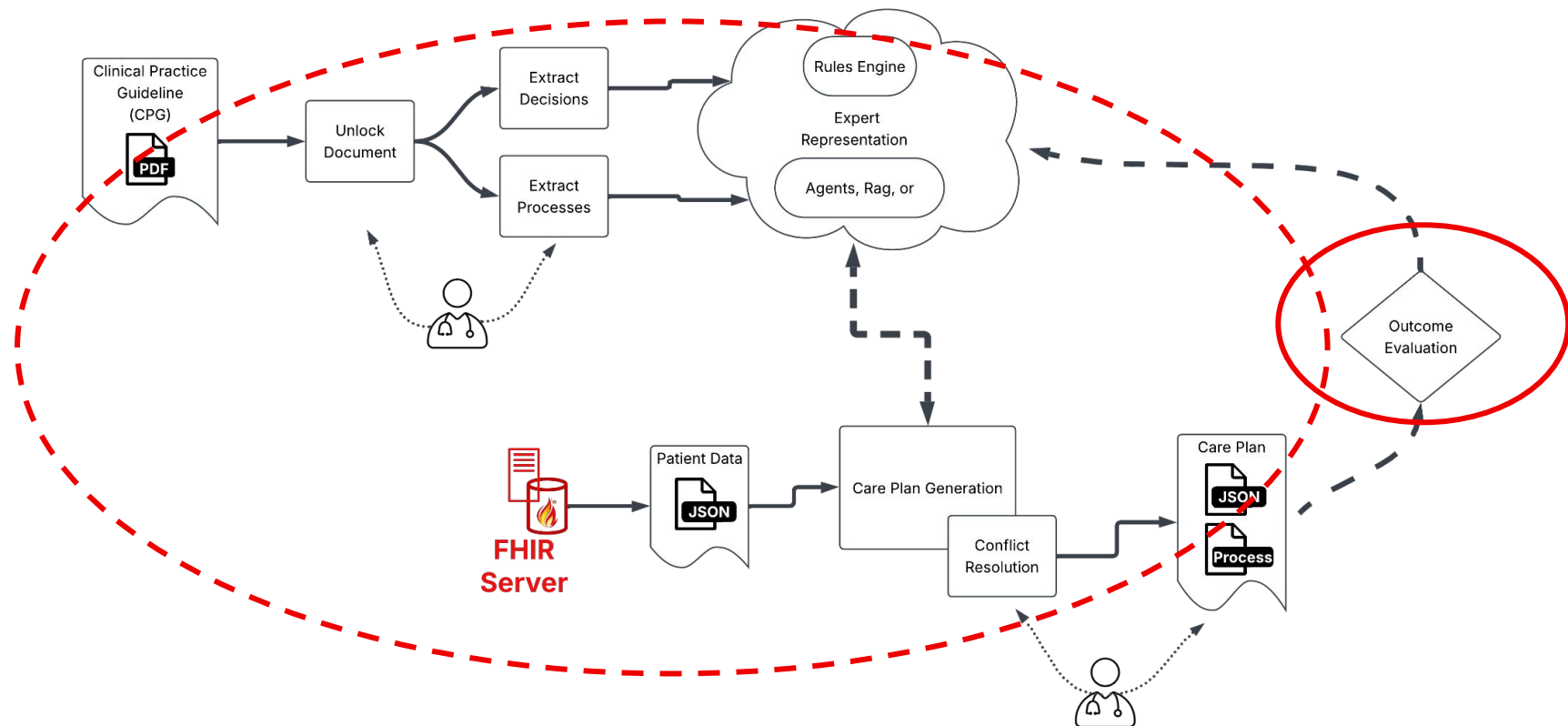
User Interfaces



	Upload Input	Review and Edit	Publish
CPG unlocking	PDF (Word and others?)	DMN and Process Representation	To care plan writer
Care Plan writing	IPS (Text and other formats?)	Conflict resolution, goals, activities (BPMN)	To FHIR server and automation engine



Future Work



- ▶ Feedback loops and learning
- ▶ Using smaller models
- ▶ Different ways of representing expert knowledge
- ▶ Importing more data formats, including converting non-FHIR to FHIR to create IPS
- ▶ Scaling all parts of the solution
- ▶ Swappable technology solutions
- ▶ ...



Thank you

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